

Thesis draft

Combining Flexible Algorithms and Latent Variable Approaches for
Baseline Covariate Adjustment in Longitudinal Models.

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1 Introduction

Researchers in behavioral and psychological research are often interested in how two variables influence each other over time. Imagine a researcher hypothesizes that a football player’s self-confidence (X) causally affects his/her match performance (Y), and vice versa. Figure 1 displays the corresponding assumed causal structure between repeated measures of X_t and Y_t . Because neither variable can be directly manipulated, the researcher relies on observational data from a random sample of football players. In this setting, psychologists have frequently used the Cross-Lagged Panel Model (CLPM) to infer causal effects (Usami et al., 2019). In a CLPM, each variable at time t is regressed on its own previous value and the previous value of the other variable. The effect of X_{t-1} on X_t is referred to as the auto-regressive parameter, whereas the effect of X_{t-1} on Y_t (and vice versa) is captured by the cross-lagged parameter. The researcher is primarily interested in the latter, interpreting the cross-lagged parameter as the causal effect of one variable on the other over time.

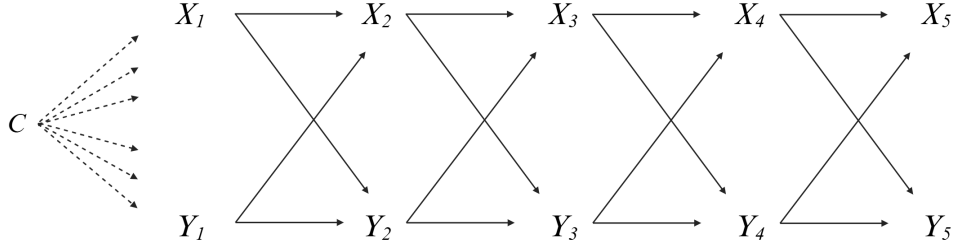


Figure 1: Reciprocal causal effects of variables X_t and Y_t , with a set of baseline covariates (C).

A key threat to any observational study are confounders: common causes of both self-confidence and performance. For example, older, more experienced players may understand their own abilities better, which increases their confidence and also leads to stronger match performance. Age and other covariates, which do not meaningfully change over the course of a study, are referred to as baseline confounders (C). These covariates are sometimes seen as stable between-person differences, whereas causal effects are assumed to unfold within persons over time Lucas (2023). Unlike baseline confounders, time-varying confounders are variables whose values can change. One example is a

player sustaining an injury during the study period. Both types of confounders lead to biased estimates of the cross-lagged parameters when not, or inappropriately controlled for. In this thesis, we consider only time-invariant confounders.

While several modeling traditions, including Marginal Structural Models [reference] and multilevel models [reference], are frequently used to estimate causal effects from longitudinal data, Structural Equation Modeling (SEM) has remained the most widely used framework in the social sciences (Tomarken and Waller, 2005). The Cross-Lagged Panel Model (CLPM) is typically estimated using this framework. Within SEM, baseline confounding is generally addressed in one of two ways. The first is direct control, in which observed baseline confounders are explicitly included in the model, typically by adding them as control variables at each measurement occasion [reference]. The second is indirect control, in which stable between-person differences are accounted for through the structure of the model itself rather than through the direct inclusion of observed confounders. One approach is to include latent variables that represent stable characteristics on which individuals may differ (e.g. age). In this context, the Random Intercepts Cross-Lagged Panel Model (RI-CLPM) (Hamaker et al., 2015) has become a widely used extension of the traditional CLPM over the last decade. A related approach is the DPM (Dishop and DeShon, 2022), which originated in econometrics and has since been adapted for estimation within the SEM framework (Allison et al., 2017). Finally, instrumental variable methods deserve brief mention, as they have also been incorporated into the SEM framework [reference]. They represent another way of addressing confounding [reference book], but lie beyond the scope of the present discussion.

With the main approaches to baseline confounding now outlined, we begin by examining the assumptions underlying methods based on direct control of baseline confounders. An obvious limitation of these approaches is that all relevant confounders must be observed, an assumption commonly referred to as no unmeasured confounding [reference]. In practice, it is rarely known if this assumption holds [reference]. Another challenge is that researchers must specify how confounders affect the variables of interest. In practice, social scientists often assume that such effects are linear. Unfortunately, relationships between variables studied in the social sciences often do not exert linear

effects (Hayes et al., 2007). In our example, age may not affect football performance in a simple linear fashion, but may instead interact with other variables, such as fitness and playing position. ~~Suppose increasing age leads to a decline in physical fitness, and physical fitness in turn influences performance. Then the effect of age on performance will partly depend on how important fitness is for that player’s role, which is likely greater for a midfielder than for a goalkeeper.~~ These interaction effects are not adequately controlled for by adding age and playing position only as linear covariates (Mize, 2019). Although some work has extended the SEM framework to accommodate non-linear relations (Muthén, 2002), these approaches have largely been developed with interpretability and hypothesis testing in mind. As a result, researchers must still specify the functional form of the confounder effects in advance. A recent approach, baseline covariate adjustment (BCA) SEM (Irmer et al., 2025) decouples control for time-invariant confounders from the estimation of the cross-lagged effects. BCA scores, residuals from regressing variables of interest on baseline covariates, replace observed variable in the SEM. A major strength of this approach is that the relationship between the variables of interest and the confounders can be flexibly modeled using machine learning (ML) techniques, such that the form of the adjustment function does not need to be assumed but can be learned from the data. This should in principle reduce model misspecification, and lead to more robust causal inference. Unfortunately, using ML techniques has its own set of challenges. ML models typically need substantially more parameters than simple linear models to capture complex functional forms. As a result, each parameter is estimated from less information, which increases the variability of the parameter estimates relative to linear models. This creates a trade-off: even if the ML approach is unbiased on average, in practice researchers are more likely to obtain estimates that deviate substantially from the true cross-lagged effect. A related limitation is that separating covariate adjustment from model estimation can lead to underestimated standard errors for the cross-lagged effects if uncertainty from the adjustment step is ignored (Freckleton, 2002). Because the SEM is fit only after adjustment, it treats the residualized data as known, rather than as having been generated by an estimated model. Furthermore, while machine learning methods may improve adjustment and reduce the need to specify the functional form in advance, this flexibility comes at a cost. Treating the confounder model as a nuisance function provides little insight into the

confounding relationship itself. Despite these concerns, machine learning methods have been successfully applied to longitudinal data (Sheetal et al., 2023), demonstrating their potential to reduce model misspecification bias. The first aim of this thesis is to investigate whether machine learning methods implemented with BCA can effectively address non-linear effects of baseline confounders within the SEM framework.

Next, we turn our attention to the assumptions underlying latent variable methods. An initial advantage of these approaches is that they do not require the functional form of stable between-person differences to be specified in advance. Whether such differences arise through a linear or non-linear process is less important, provided that they remain stable over time. This points to the central statistical assumption of latent-variable control: that the effects of baseline confounders do not vary over time, i.e. that they are stationary (Murayama and Gfrörer, 2024). However, baseline confounders, while time-invariant themselves, may have effects that vary over time. In the example study, a rule change by the national governing body of football may encourage high-intensity play. This shift benefits younger, fitter players, reversing the effect of age on performance, changing it from positive before the change, to negative afterward. When the effects of time-invariant confounders vary over time, bias in the cross-lagged and auto-regressive parameters may persist [reference]. One proposed solution is to free the factor loadings of the latent variable, thereby allowing the latent factor representing baseline confounders to have time-varying effects. However, closer inspection suggests that this strategy is effective only if the relative strength of the unobserved confounders remains constant over time [reference in prep]. That is, a confounder that is stronger than another at one time point must remain stronger at later time points as well. In practice, this assumption cannot be tested.

A natural possibility is to combine both approaches. Researchers may first adjust directly for observed confounders, either linearly or with more flexible methods, and then use latent variables to capture any remaining unobserved baseline confounding. This does not eliminate all risk of bias: bias may still persist if unobserved confounders exert time-varying effects, or if the functional form of the observed confounders is misspecified. Nevertheless, such a combined strategy offers an

important benefit. Once observed confounders have been adjusted for directly, their effects can no longer bias the estimates simply because they vary over time; at the same time, latent variables may still account, at least partially, for confounding due to omitted baseline factors. A stronger objection to such an omnibus approach is the amount of data needed to estimate models reliably. For instance, the RI-CLPM is known to encounter convergence issues when sample sizes are too small, often below roughly 300 individuals [reference]. As noted earlier, machine learning methods may reduce bias, but often do so at the cost of higher variance, yielding estimates that are less stable in practice. When these flexible adjustment methods are combined with more demanding longitudinal models such as the RI-CLPM or DPM, sample size may become the main practical constraint. The second aim of this thesis is to assess whether combining flexible models with methods for unobserved confounding can effectively reduce bias in settings where some confounders are unobserved and the effects of confounders vary over time. The second aim of this thesis is to examine whether combining flexible adjustment methods with approaches for unobserved confounding can effectively address confounding when not all confounders are observed and their effects vary over time. This will be evaluated across a range of sample sizes, to determine how sensitive these combined approaches are to the amount of available data. Finally, we will also examine how the combined approach performs in settings where the assumptions of stationarity and unobserved confounding are not violated. In essence, this addresses the question of how much reduction in estimator variance can be achieved by relying on these assumptions.

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